

Generalized Additive Model for Council District V (2023 - 2025 Sales)

Elliot Dean - Program Analyst IV
Office of the Chief Financial Officer | Office of the Assessor
City of Detroit

7 April 2026

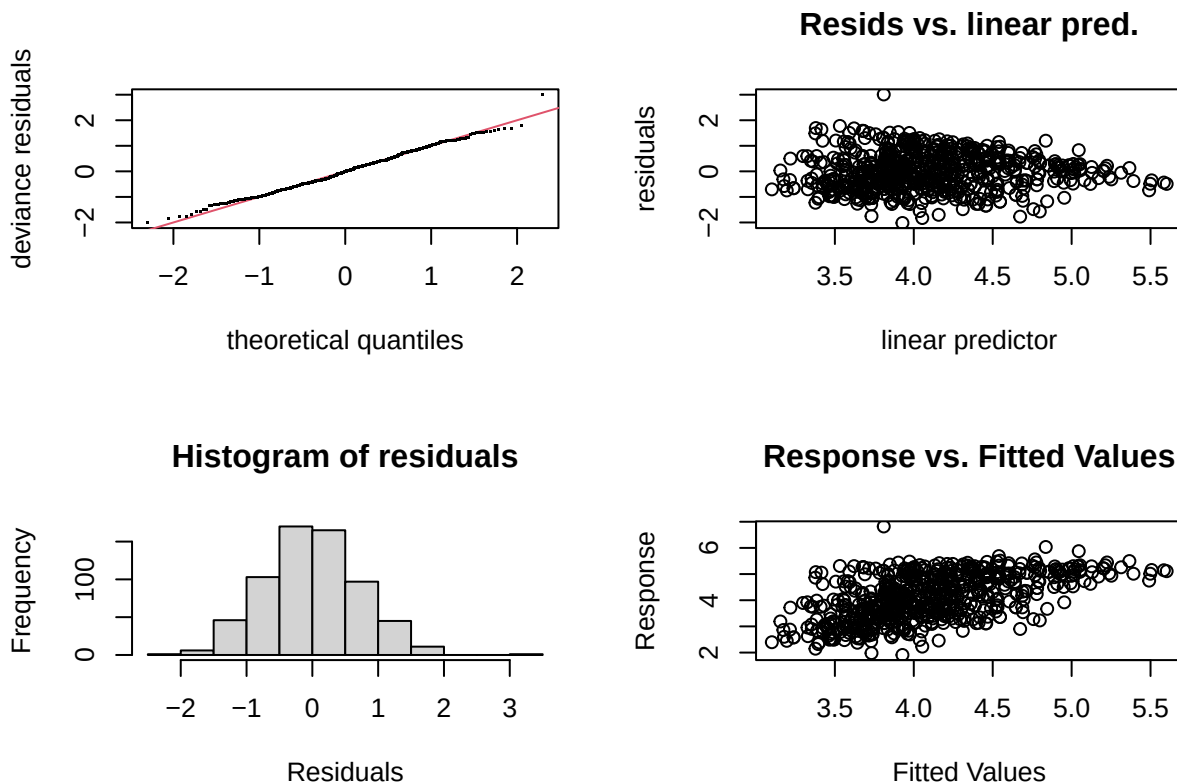
Contents

Model I	1
Model II	3
Model III	5
Model IV	8

Model I

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_sppsf ~ arterial + single_family1story + half_duplex1story +
##   half_duplex2story + smallest + small + large + largest +
##   two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##   qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##   crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##   hot_water + elec_baseboard + wall + central_air + s(months,
##   bs = "cs", k = 10)
##
## Parametric coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.056040   0.111153  36.491 < 2e-16 ***
## arterial       -0.157134   0.083710  -1.877  0.060972 .
## single_family1story 0.171906   0.070209   2.448  0.014623 *
## half_duplex1story  0.000000   0.000000    NaN      NaN
## half_duplex2story -0.224276   0.734765  -0.305  0.760290
## smallest        0.122358   0.293679   0.417  0.677087
## small          -0.218969   0.224458  -0.976  0.329674
## large          -0.100532   0.107598  -0.934  0.350499
## largest         0.029187   0.132679   0.220  0.825960
## two_plus_baths   0.108607   0.107190   1.013  0.311351
## powder_room      0.006741   0.072589   0.093  0.926043
## good             0.787077   0.111709   7.046  4.95e-12 ***
## fair            -0.367169   0.067206  -5.463  6.79e-08 ***
## poor            -0.401962   0.101833  -3.947  8.82e-05 ***
```

```
## very_poor_unsound    -0.701075    0.210564   -3.330 0.000922 ***
## qabove_avg           0.517419    0.105760    4.892 1.27e-06 ***
## qbelow_avg          -0.136770    0.129958   -1.052 0.293022
## crawl                0.047006    0.170983    0.275 0.783472
## slab                 0.082289    0.319726    0.257 0.796977
## fireplace           -0.003840    0.072650   -0.053 0.957860
## hot_water            0.076311    0.069406    1.099 0.271986
## elec_baseboard       0.000000    0.000000     NaN    NaN
## wall                 0.000000    0.000000     NaN    NaN
## central_air          0.035600    0.139687    0.255 0.798921
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df    F  p-value
## s(sresb_yearbuilt) 4.9388     9 2.260 0.000241 ***
## s(ln_garagespaces) 0.5829     3 0.370 0.146140
## s(months)          2.3040     9 3.208 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Rank: 42/45
## R-sq.(adj) =  0.277   Deviance explained = 30.8%
## -REML = 740.37   Scale est. = 0.52693    n = 645
```



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-2.07195e-05,6.839048e-07]
## (score 740.3722 & scale 0.5269289).
```

```
## Hessian positive definite, eigenvalue range [0.1595309,310.5242].
## Model rank = 42 / 45
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##          k'    edf k-index p-value
## s(sresb_yearbuilt) 9.000 4.939    0.87 <2e-16 ***
## s(ln_garagespaces) 3.000 0.583    0.78 <2e-16 ***
## s(months)          9.000 2.304    0.91  0.025 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio prd cod prb
##   <int>      <dbl>      <dbl>      <dbl> <dbl> <dbl> <dbl>
## 1   645    0.000438    0.000621    0.000340  1.83  82.2 -1.01
```

Table 1: District 5 LN_SPPSF NLM I Assessing Coefficients

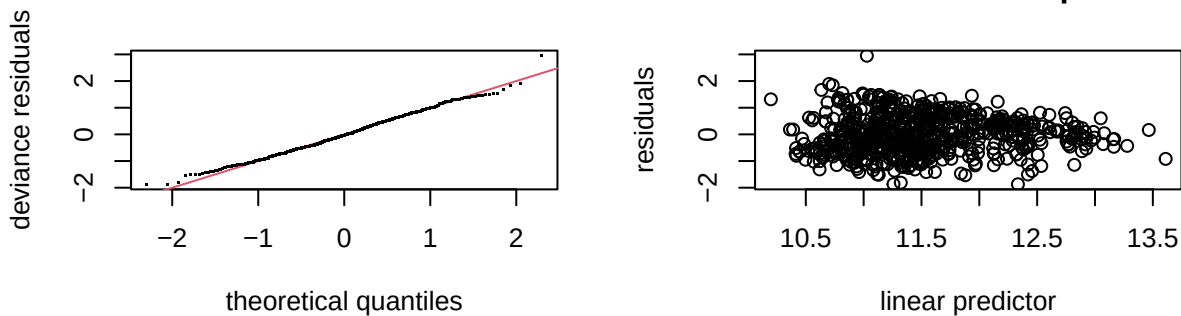
n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
645	0.0005611	0.0008045	0.0004445	1.809726	82.14481	-0.0005536

Model II

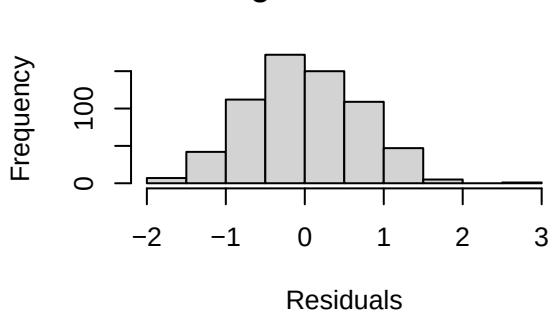
```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##   bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##   half_duplex2story + smallest + small + large + largest +
##   two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##   qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##   crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##   hot_water + elec_baseboard + wall + central_air + s(months,
##   bs = "cs", k = 10)
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   11.36740    0.14776  76.933 < 2e-16 ***
## arterial      -0.15559    0.08393  -1.854  0.064265 .
## single_family1story 0.13539    0.07107   1.905  0.057248 .
## half_duplex1story  0.00000    0.00000    NaN      NaN
## half_duplex2story -0.05423    0.73500  -0.074  0.941210
## smallest      -0.32730    0.34172  -0.958  0.338530
## small         -0.31654    0.23317  -1.358  0.175118
## large          0.01812    0.12149   0.149  0.881483
## largest        0.25249    0.17194   1.468  0.142484
## two_plus_baths  0.18168    0.12673   1.434  0.152217
```

```
## powder_room      0.04084    0.07565    0.540 0.589536
## good             0.82223    0.11192    7.347 6.47e-13 ***
## fair            -0.35162    0.06672   -5.270 1.89e-07 ***
## poor            -0.38295    0.10138   -3.777 0.000174 ***
## very_poor_unsound -0.73018    0.20917   -3.491 0.000516 ***
## qabove_avg       0.45636    0.11458    3.983 7.62e-05 ***
## qbelow_avg      -0.13630    0.12972   -1.051 0.293786
## crawl            0.05567    0.16548    0.336 0.736687
## slab             0.06199    0.31921    0.194 0.846086
## fireplace       -0.09791    0.06951   -1.409 0.159424
## hot_water        0.06656    0.06969    0.955 0.339891
## elec_baseboard   0.00000    0.00000     NaN     NaN
## wall             0.00000    0.00000     NaN     NaN
## central_air      -0.10779    0.13506   -0.798 0.425118
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F  p-value
## s(ln_landratio)  3.78898      9 2.532 2.91e-05 ***
## s(ln_base_bldgsize_ratio) 1.02948      9 0.423  0.0195 *
## s(sresb_yearbuilt)  0.82575      9 0.262  0.0798 .
## s(ln_garagespaces)  0.00125      3 0.000  0.5626
## s(months)          2.51634      9 4.196 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Rank: 60/63
## R-sq.(adj) =  0.424   Deviance explained = 44.9%
## -REML = 733.22   Scale est. = 0.52581    n = 645
```

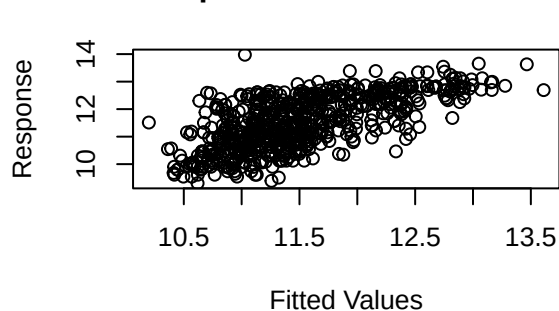
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 11 iterations.
## Gradient range [-0.0003691536,0.0002544998]
## (score 733.2155 & scale 0.5258108).
## Hessian positive definite, eigenvalue range [0.0003690775,310.5178].
## Model rank = 60 / 63
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##              k'      edf k-index p-value
## s(ln_landratio)    9.00000 3.78898    0.81 <2e-16 ***
## s(ln_base_bldgsize_ratio) 9.00000 1.02948    0.97 0.190
## s(sresb_yearbuilt)    9.00000 0.82575    0.87 0.005 **
## s(ln_garagespaces)    3.00000 0.00125    0.78 <2e-16 ***
## s(months)            9.00000 2.51634    0.91 0.015 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio prd   cod   prb
##   <int>      <dbl>      <dbl>      <dbl> <dbl> <dbl> <dbl>
## 1   645        0.781        0.979        0.630  1.55  66.6 -0.607
```

Table 2: District 5 LN_PRICE (No ECFs) NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
645	1.029011	1.270461	0.824281	1.541296	64.02327	-0.6909933

Model III

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##   bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##   half_duplex2story + smallest + small + large + largest +
##   two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##   qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##   crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##   hot_water + elec_baseboard + wall + central_air + s(months,
##   bs = "cs", k = 10) + ecf
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    10.956007   0.146427   74.822 < 2e-16 ***
## arterial        -0.141200   0.081269   -1.737 0.082837 .
## single_family1story 0.136682   0.065549    2.085 0.037484 *
```

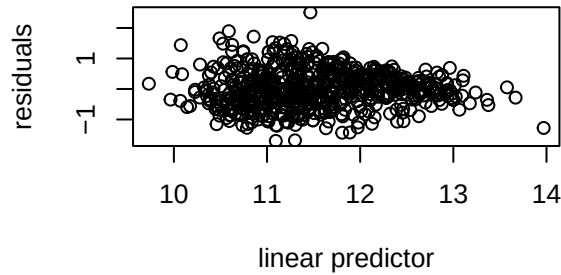
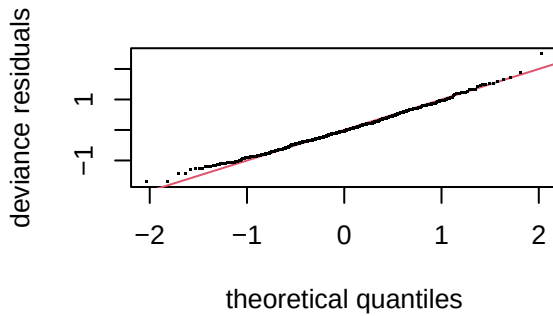
```

## half_duplex1story    0.000000    0.000000    NaN    NaN
## half_duplex2story   -0.670811    0.693906   -0.967  0.334085
## smallest            -0.287389    0.298095   -0.964  0.335400
## small              -0.260028    0.205762   -1.264  0.206831
## large               0.001244    0.105418    0.012  0.990592
## largest            0.142945    0.145831    0.980  0.327385
## two_plus_baths      0.245183    0.111951    2.190  0.028911 *
## powder_room         0.064687    0.068443    0.945  0.344991
## good                0.788198    0.101330    7.779  3.33e-14 ***
## fair               -0.370792    0.060797   -6.099  1.94e-09 ***
## poor               -0.510632    0.092665   -5.510  5.37e-08 ***
## very_poor_unsound  -0.898923    0.209847   -4.284  2.15e-05 ***
## qabove_avg          0.209909    0.134147    1.565  0.118179
## qbelow_avg         -0.219701    0.127697   -1.720  0.085872 .
## crawl              0.201447    0.155832    1.293  0.196618
## slab               -0.135266    0.321267   -0.421  0.673881
## fireplace           0.029950    0.076399    0.392  0.695188
## hot_water           0.071046    0.063741    1.115  0.265476
## elec_baseboard      0.000000    0.000000    NaN    NaN
## wall               0.000000    0.000000    NaN    NaN
## central_air         0.016382    0.124888    0.131  0.895682
## ecf5R502            0.150272    0.098362    1.528  0.127117
## ecf5R503            1.085879    0.252285    4.304  1.96e-05 ***
## ecf5R504            0.804555    0.161663    4.977  8.51e-07 ***
## ecf5R505            0.466289    0.347771    1.341  0.180508
## ecf5R506            0.749508    0.300654    2.493  0.012945 *
## ecf5R507            0.329104    0.176985    1.860  0.063457 .
## ecf5R508            0.941969    0.412489    2.284  0.022751 *
## ecf5R509            0.954320    0.231119    4.129  4.17e-05 ***
## ecf5R510            0.218715    0.150354    1.455  0.146297
## ecf5R511            0.824856    0.243501    3.387  0.000753 ***
## ecf5R512            0.879753    0.225885    3.895  0.000110 ***
## ecf5R513            0.628984    0.317778    1.979  0.048248 *
## ecf5R514            0.890348    0.125604    7.089  3.92e-12 ***
## ecf5R516            1.161833    0.744586    1.560  0.119212
## ecf5R517            1.545923    0.303274    5.097  4.65e-07 ***
## ecf5R518            0.664093    0.407015    1.632  0.103298
## ecf5R521            0.720287    0.178509    4.035  6.18e-05 ***
## ecf5R523           -0.113702    0.199630   -0.570  0.569191
## ecf5R524            2.441943    0.528623    4.619  4.73e-06 ***
## ecf5R528            0.236753    0.152413    1.553  0.120875
## ecf5R529            0.920680    0.127726    7.208  1.76e-12 ***
## ecf5R530            1.129417    0.275836    4.095  4.83e-05 ***
## ecf5R531            0.468480    0.144014    3.253  0.001208 **
## ecf5R532            0.997519    0.345026    2.891  0.003981 **
## ecf5R533           -0.111807    0.159076   -0.703  0.482428
## ecf5R534            0.176749    0.144773    1.221  0.222628
## ecf5R535            0.687506    0.499082    1.378  0.168871
## ecf5R538            0.295192    0.212210    1.391  0.164743
## ecf5R539            0.451461    0.282015    1.601  0.109951
## ecfCGRS6            0.284090    0.669486    0.424  0.671473
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(ln_landratio)      2.8262      9 1.091 0.00879 **
## s(ln_base_bldgsize_ratio) 0.6546      9 0.158 0.11998
## s(sresb_yearbuilt)    0.8254      9 0.384 0.03461 *

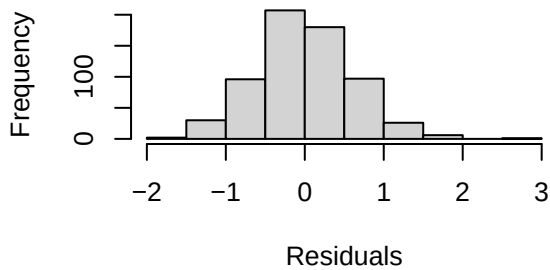
```

```
## s(ln_garagespaces)      0.9605      3 1.162 0.03429 *
## s(months)               3.2186      9 5.196 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Rank: 90/93
## R-sq.(adj) =  0.55   Deviance explained = 59.1%
## -REML = 659.29   Scale est. = 0.41122   n = 645
```

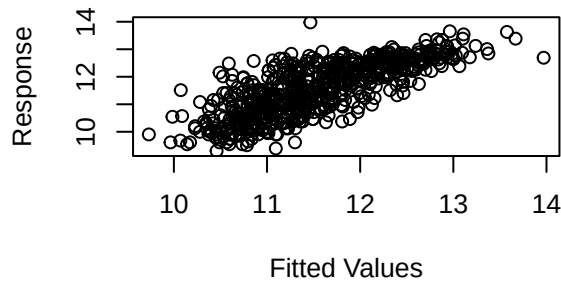
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 8 iterations.
## Gradient range [-2.027147e-05,7.879562e-07]
## (score 659.2885 & scale 0.411216).
## Hessian positive definite, eigenvalue range [0.1867237,295.5173].
## Model rank =  90 / 93
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##           k'   edf k-index p-value
## s(ln_landratio)      9.000 2.826   0.89 <2e-16 ***
## s(ln_base_bldgsize_ratio) 9.000 0.655   1.01   0.57
## s(sresb_yearbuilt)      9.000 0.825   1.00   0.52
## s(ln_garagespaces)      3.000 0.960   0.89 <2e-16 ***
## s(months)              9.000 3.219   1.01   0.49
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## # A tibble: 1 x 7
```

```
##      n median_ratio mean_ratio wmean_ratio   prd   cod   prb
##   <int>      <dbl>      <dbl>      <dbl> <dbl> <dbl> <dbl>
##  1    645      0.769      0.921      0.670  1.37  55.9 -0.386
```

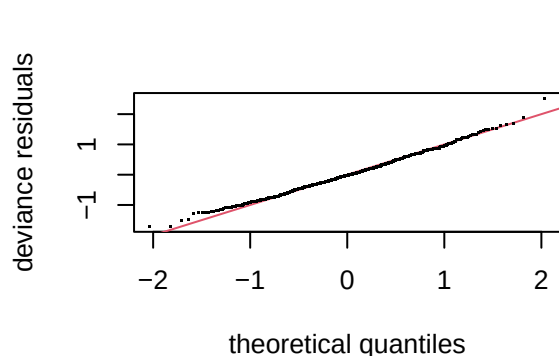
Model IV

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_tasp ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##      bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##      half_duplex2story + smallest + small + large + largest +
##      two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##      qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##      crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##      hot_water + elec_baseboard + wall + central_air + ecf + s(months,
##      bs = "cs", k = 10)
##
## Parametric coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    11.20508    0.14271   78.518 < 2e-16 ***
## arterial       -0.13937    0.08132   -1.714 0.087079 .
## single_family1story 0.16345    0.05656    2.890 0.003999 **
## half_duplex1story  0.00000    0.00000     NaN      NaN
## half_duplex2story -0.69813    0.69410   -1.006 0.314922
## smallest       -0.27909    0.29524   -0.945 0.344897
## small          -0.27976    0.20518   -1.363 0.173265
## large           0.00740    0.10437    0.071 0.943498
## largest         0.15341    0.14338    1.070 0.285050
## two_plus_baths   0.25902    0.11069    2.340 0.019611 *
## powder_room      0.06977    0.06829    1.022 0.307395
## good             0.78073    0.10080    7.745 4.18e-14 ***
## fair            -0.36892    0.06061   -6.086 2.08e-09 ***
## poor            -0.50591    0.09253   -5.468 6.74e-08 ***
## very_poor_unsound -0.89478    0.20911   -4.279 2.19e-05 ***
## qabove_avg       0.20464    0.13388    1.528 0.126926
## qbelow_avg      -0.21444    0.12747   -1.682 0.093038 .
## crawl           0.19782    0.15589    1.269 0.204956
## slab            -0.15063    0.32095   -0.469 0.639001
## fireplace        0.02975    0.07650    0.389 0.697494
## hot_water        0.06932    0.06320    1.097 0.273155
## elec_baseboard   0.00000    0.00000     NaN      NaN
## wall            0.00000    0.00000     NaN      NaN
## central_air      0.01967    0.12502    0.157 0.875022
## ecf5R502         0.14948    0.09836    1.520 0.129144
## ecf5R503         1.10444    0.25046    4.410 1.23e-05 ***
## ecf5R504         0.81144    0.16082    5.045 6.04e-07 ***
## ecf5R505         0.47723    0.34636    1.378 0.168774
## ecf5R506         0.75178    0.30037    2.503 0.012589 *
## ecf5R507         0.34552    0.17676    1.955 0.051089 .
## ecf5R508         0.96612    0.41202    2.345 0.019366 *
## ecf5R509         0.97144    0.23056    4.213 2.91e-05 ***
## ecf5R510         0.21921    0.15013    1.460 0.144779
## ecf5R511         0.82274    0.24375    3.375 0.000786 ***
```

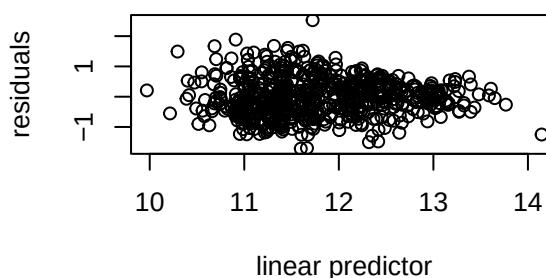
```

## ecf5R512      0.89740    0.22568    3.976 7.87e-05 ***
## ecf5R513      0.65055    0.31758    2.048 0.040961 *
## ecf5R514      0.89040    0.12558    7.090 3.85e-12 ***
## ecf5R516      1.12711    0.74428    1.514 0.130470
## ecf5R517      1.53725    0.30339    5.067 5.42e-07 ***
## ecf5R518      0.70402    0.40666    1.731 0.083932 .
## ecf5R521      0.70803    0.17796    3.979 7.79e-05 ***
## ecf5R523     -0.11885    0.19965   -0.595 0.551875
## ecf5R524      2.45372    0.52839    4.644 4.22e-06 ***
## ecf5R528      0.23123    0.15215    1.520 0.129109
## ecf5R529      0.91923    0.12764    7.202 1.82e-12 ***
## ecf5R530      1.12417    0.27596    4.074 5.26e-05 ***
## ecf5R531      0.46390    0.14390    3.224 0.001335 **
## ecf5R532      1.01632    0.34358    2.958 0.003221 **
## ecf5R533     -0.10720    0.15929   -0.673 0.501219
## ecf5R534      0.18937    0.14455    1.310 0.190660
## ecf5R535      0.72874    0.49822    1.463 0.144088
## ecf5R538      0.28443    0.21233    1.340 0.180909
## ecf5R539      0.41937    0.28162    1.489 0.136990
## ecfCGRS6      0.30118    0.67029    0.449 0.653358
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##              edf Ref.df      F p-value
## s(ln_landratio)      2.7292415      9 1.012  0.0118 *
## s(ln_base_bldgsize_ratio) 0.5938312      9 0.132  0.1404
## s(sresb_yearbuilt)      0.8404782      9 0.408  0.0304 *
## s(ln_garagespaces)      0.9698763      3 1.208  0.0327 *
## s(months)              0.0004109      9 0.000  0.6633
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Rank: 90/93
## R-sq.(adj) =  0.536   Deviance explained = 57.6%
## -REML = 655.51   Scale est. = 0.41293   n = 645

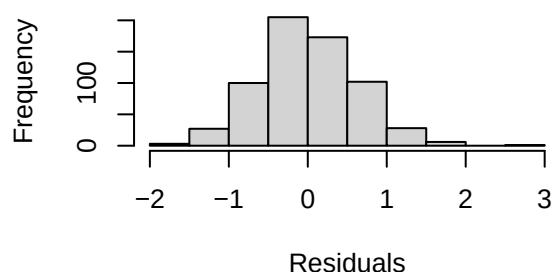
```



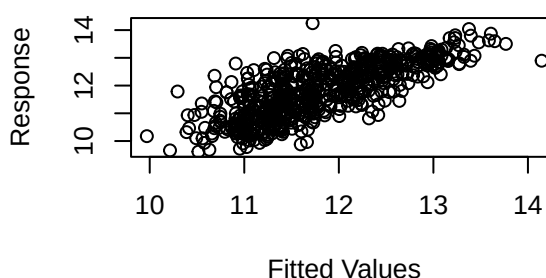
Resids vs. linear pred.



Histogram of residuals



Response vs. Fitted Values



```
##
## Method: REML   Optimizer: outer newton
## full convergence after 9 iterations.
## Gradient range [-0.0001498682,5.559484e-05]
## (score 655.515 & scale 0.4129274).
## Hessian positive definite, eigenvalue range [0.0001498284,295.5079].
## Model rank = 90 / 93
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##               k'      edf k-index p-value
## s(ln_landratio)  9.000000 2.729241   0.89 <2e-16 ***
## s(ln_base_bldgsize_ratio) 9.000000 0.593831   1.01   0.56
## s(sresb_yearbuilt)  9.000000 0.840478   1.00   0.49
## s(ln_garagespaces)  3.000000 0.969876   0.89   0.01 **
## s(months)          9.000000 0.000411   1.00   0.46
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio prd   cod   prb
##   <int>      <dbl>      <dbl>      <dbl> <dbl> <dbl> <dbl>
## 1   645         1.02         1.20         0.870 1.38  52.7 -0.390
```